



CAREERNOVA-AI

AN AI-POWERED RESUME TRANSFORMATION &
PORTFOLIO GENERATION PLATFORM

Subtitle

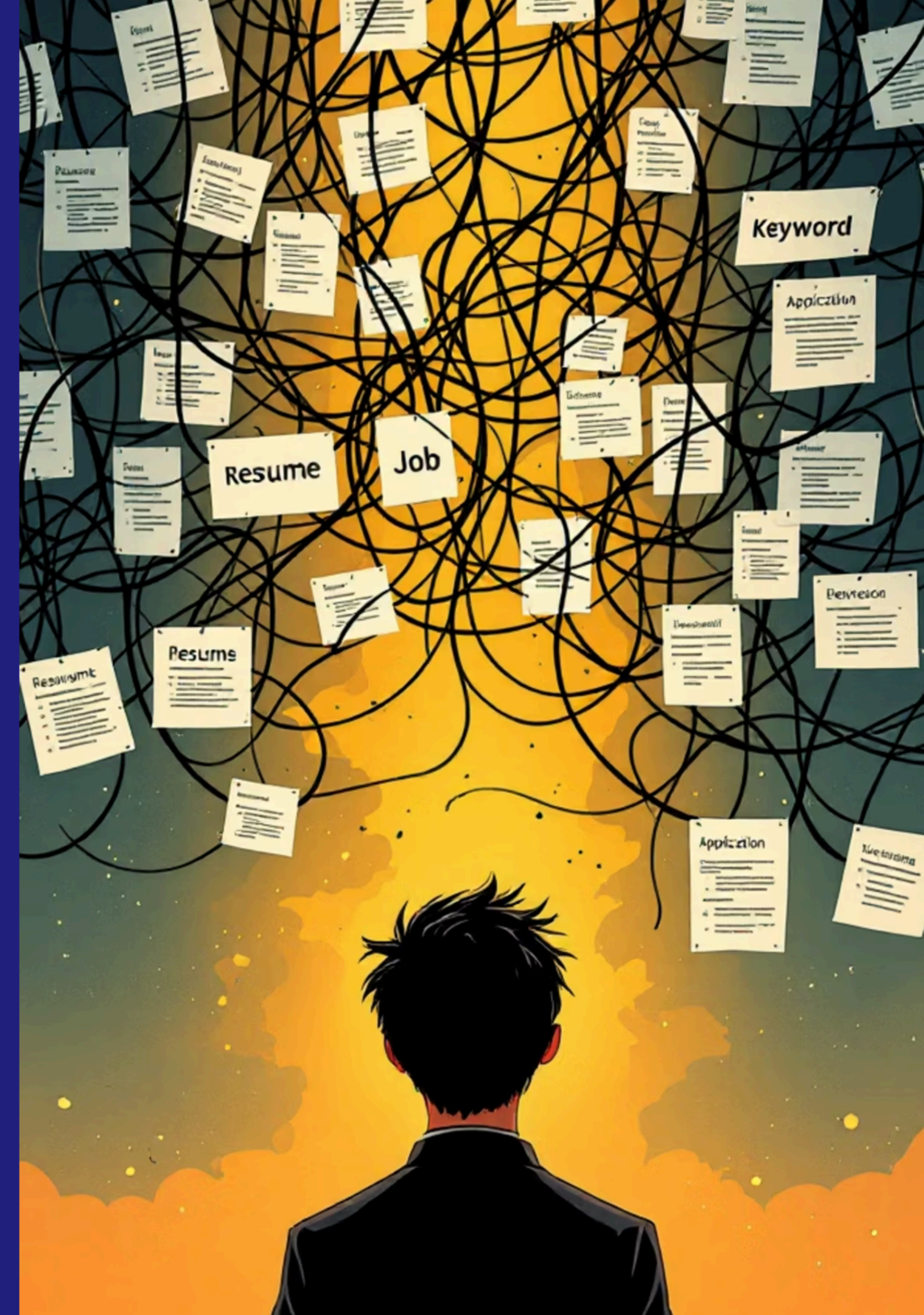
Transforming Traditional Resumes into Dynamic
Portfolios & ATS-Optimized Professional Assets.

  www.careernova.ai 



PROBLEM STATEMENT & INDUSTRY GAP

- Many resumes fail to pass Applicant Tracking Systems (ATS) due to poor optimization and formatting.
- Applicants struggle to secure suitable jobs because resumes are not aligned with job descriptions.
- Candidates lack guidance on improving ATS scores and increasing shortlisting chances.
- Students face challenges in creating professional portfolios.
- Resume customization for each role is time-consuming and inefficient.
- Applicants are unsure about potential interview questions based on their resume.



Proposed Solution

Intelligent Resume Transformation,
and ATS Optimization & Career Recommendation Platform



CareerNova AI

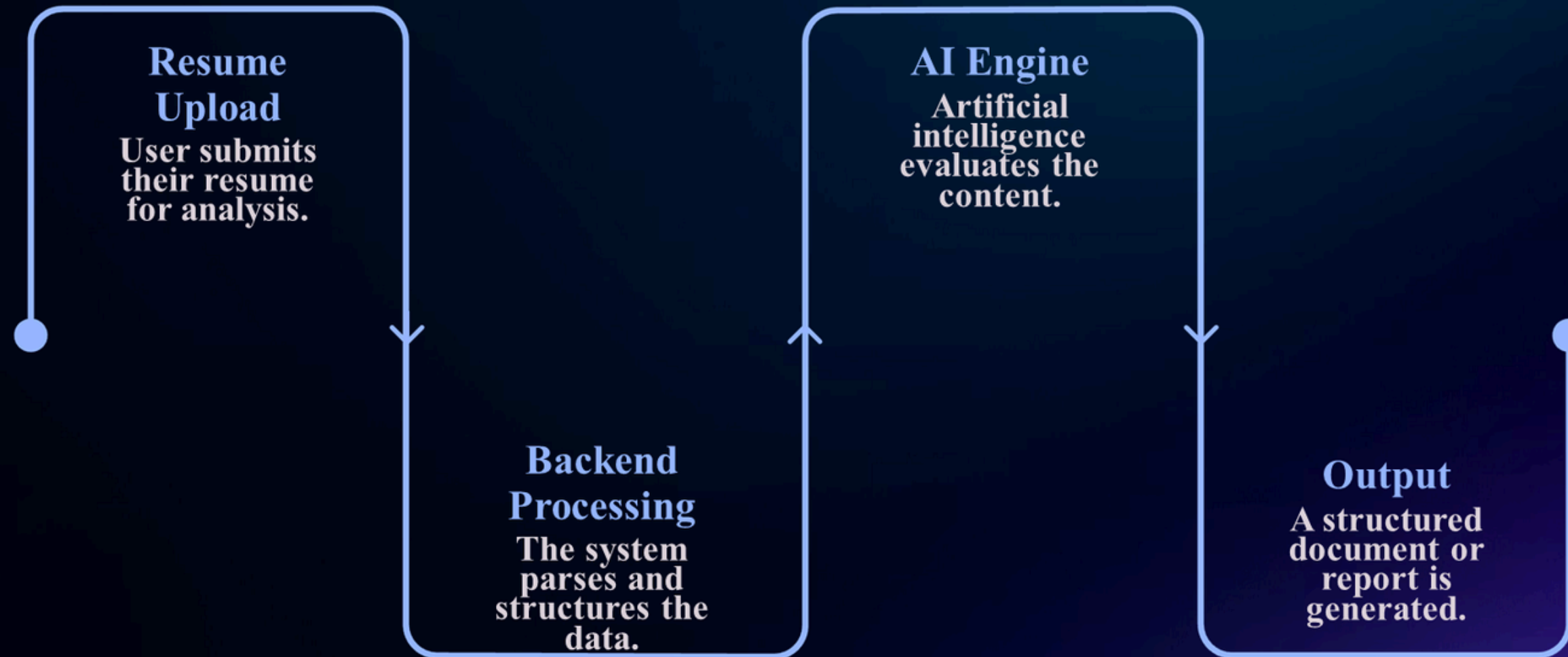
PROPOSED SOLUTION – CAREERNOVA AI

Intelligent Resume Transformation & Career Optimization Platform

- | AI-Powered Resume Analysis & Structuring
- | ATS Score Evaluation with Smart Improvement Suggestions
- | Automatic Portfolio Generation
- | ATS-Optimized Resume Creation
- | Resume-Based Job Recommendations
- | Personalized Interview Question Suggestions
- | Chatbot-Assisted Career Guidance
- | Multi-Format Export (PPTX & DOCX)

System Architecture & Workflow

Intelligent Processing Pipeline



- **Resume Upload (PDF/DOCX Input Layer)**
- **Backend Processing & Resume Parsing Engine**
- **Secure Data Storage – Neon PostgreSQL**
- **AI Engine (LangChain + OpenRouter Integration)**
- **ATS Scoring & Match Calculation Module**
- **Portfolio & Resume Generation Engine**
- **Document Export & Cloud Deployment (Render)**

Secure • Scalable • Cloud-Based

🔧 Technology Stack Overview

🌐 Frontend

- HTML5, CSS3, JavaScript
- Tailwind CSS (Glass UI, Light/Dark Mode)

⚙️ Backend

- FastAPI (REST API Framework)
- Node.js / Python
- Resume Parsing & File Handling

🧠 AI & LLM Integration

- LangChain (LLM Orchestration)
- OpenRouter API (LLM Access)
- Resume Analysis & ATS Optimization
- Job Recommendation & Match Scoring
- Interview Question Generator

🗄️ Database

- Neon PostgreSQL (Serverless Cloud Database)

📄 Document Generation

- PPTX Portfolio Generator
- DOCX ATS-Optimized Resume

☁️ Deployment

- GitHub (Version Control)
- Render (Cloud Hosting)



KEY FEATURES

AI-Powered Resume Analysis

ATS Score Calculation & Optimization Suggestions

ATS-Friendly Resume Builder

One-Click Portfolio Generation

Job Recommendation & Match Scoring

Personalized Interview Question Suggestions

Resume Improvement Guidance

Chatbot-Assisted Career Support

PPTX Portfolio & DOCX Resume Export

Secure Cloud-Based Processing

UI/UX & Design Innovation

- 3D Glass effect Interface
- Modern SaaS-Style Layout
- Light & Dark Theme System
- Fully Responsive Design (Mobile + Desktop)
- Smooth Animations & Micro-Interactions
- Clean, Professional Dashboard Experience



FUTURE SCOPE & ENHANCEMENTS

- **Multi-Theme & Industry-Specific Portfolio Templates**
- **Personal Portfolio Website Hosting (Custom Domain Support)**
- **Multi-Language Resume Generation (Global Ready)**
- **Instant Job Vacancy Detection & Faster Job Recommendations**
- **Automated Email Notification System (Gmail Alerts for New Matching Jobs)**
- **AI-Based Mock Interview Simulation**
- **Coding Interview Practice (Resume-Based Question Generation)**

THANK YOU

